

Discovering the Versatility of User Mode Linux

User Mode Linux or 'Linux in Linux' allows you to run Linux within itself. This gives users a powerful means of doing all sorts of things like debugging kernels, studying processes, etc. This article is a guide to building User Mode Linux (UML) and running multiple UML instances on the same system.



Today, Linux is one of the most widely used kernels for personal computers, embedded systems, etc, and hence everybody wants to understand and learn this important piece of code for various reasons such as educational purposes, for projects and to improve one's career options. Linux is a multi-processor, pre-emptive kernel with loadable modules, and has all the features that any modern operating system kernel has, such as user space processes, a file system, paged memory management, device drivers, kernel threads, etc.

Kernel code is too vast to understand and debug at one go. So one of the ways to study the Linux kernel is to browse the source code and understand the flow. Let's suppose you want to crash the running kernel to study debugging or change the way the user-space of your kernel boots by changing startup scripts. Such changes could create havoc on your machine and can't be easily done on your personal computer. The other option is to buy some small hardware that can boot Linux. You could then build a kernel for it and play around with it; but again, the problem is that even that hardware could get

damaged and cost you yet more money to repair. If you want to just learn the kernel's features and want to avoid the above procedures, one option is User Mode Linux.

User Mode Linux is 'Linux on Linux'. By adopting this method, you could run a Linux instance on your host machine like any other user space process such as the 'ls' program. User Mode Linux is a way of compiling Linux as a user space program, which can be executed on a terminal. With this, you can run multiple virtual Linux kernels as processes. You can crash each of them without doing any harm to your host machine. Besides, you can debug your virtual Linux in GDB. With this method, you can learn about the different parts of the kernel by debugging them when it is running.

So let's explore the way in which you can build User Mode Linux.

Building User Mode Linux (UML)

User Mode Linux (UML) has full support in the official kernel release. It can be built by going through the following steps.

1. Download the latest kernel from <https://www.kernel.org>.